HDOC Day 42

Question: Write a menu-driven C program using separate functions to remove all vowels from a string, find the first repeating lowercase alphabet, and check whether two strings are anagrams.

C Program:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
#define SIZE 200

void removeVowels(const char str[], char result[]) {
    int i, j = 0;
    for (i = 0; str[i] != '\0'; i++) {
        char c = tolower((unsigned char)str[i]);
        if (c != 'a' && c != 'e' && c != 'i' &&
            c != 'o' && c != 'u') result[j++] = str[i];
    }
    result[j] = '\0';
}

char firstRepeatingLowercase(const char str[]) {
    int seen[26] = {0}, i;
    for (i = 0; str[i] != '\0'; i++) {
        if (str[i] >= 'a' && str[i] <= 'z') {
            int index = str[i] - 'a';
            if (seen[index]) return str[i];
            seen[index] = 1;
        }
    }
    return '\0';
}

int areAnagrams(const char a[], const char b[]) {
    int freq[26] = {0}, i;
    for (i = 0; a[i]; i++) {
        unsigned char c = (unsigned char)a[i];
        if (isalpha(c)) freq[tolower(c) - 'a']++;
    }
    for (i = 0; b[i]; i++) {
        unsigned char c = (unsigned char)b[i];
        if (isalpha(c)) freq[tolower(c) - 'a']--;
    }
    for (i = 0; i < 26; i++) if (freq[i] != 0) return 0;
    return 1;
}

void readString(char s[]) {
    fgets(s, SIZE, stdin);
    s[strcspn(s, "\n")] = '\0';
}

int main(void) {
    char a[SIZE], b[SIZE], result[SIZE], repeated;
    int choice;
    do {
        printf("\n1.Remove vowels\n2.First repeating lowercase letter\n");
        printf("3.Check anagrams\n4.Exit\nEnter choice: ");
        if (scanf("%d", &choice) != 1) break;
        while (getchar() != '\n');
        switch (choice) {
            case 1:
                printf("Enter string: "); readString(a);
                removeVowels(a, result);
                printf("Result: %s\n", result); break;
            case 2:
                printf("Enter string: "); readString(a);
```

```
                repeated = firstRepeatingLowercase(a);
                if (repeated) printf("First repeating: %c\n", repeated);
                else printf("No repeating lowercase letter\n");
                break;
            case 3:
                printf("First string: "); readString(a);
                printf("Second string: "); readString(b);
                printf(areAnagrams(a, b) ? "Anagrams\n" : "Not anagrams\n");
                break;
            case 4: printf("Exit\n"); break;
            default: printf("Invalid choice\n");
        }
    } while (choice != 4);
    return 0;
}
```

Sample Output:

```
1.Remove vowels
2.First repeating lowercase letter
3.Check anagrams
4.Exit
Enter choice: 1
Enter string: Education
Result: dctn
Enter choice: 2
Enter string: programming
First repeating: r
Enter choice: 3
First string: listen
Second string: silent
Anagrams
```